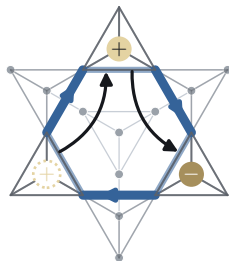


i.a) contractible hexagon

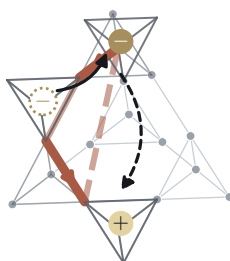
$$g_6 = 12|J_{\pm}|^3/J_{zz}^2$$



$$\mathbf{q}_\gamma = (0, 0, 0)$$

i.b) winding four-loop

$$g_4 = 4J_{\pm}^2/J_{zz}$$



$$\mathbf{q}_\gamma = (0, 0, -1)$$

